

Behavioral Research Workflow Workbook

A hands-on companion to the workshop deck -- setup, prompt templates, and troubleshooting

Behavioral Research Workshop

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Quick Start: the whole flow in one page

If you only read one page before opening a terminal, read this one. Everything here is expanded later.

1. **Set up once:** VS Code + Claude Code, Python venv, git/GitHub, pandoc (Section 1).
2. **Scaffold a project:** the folder layout in Section 2 -- one projects/<slug>/ per study.

3. **Stage 3 (design):** give Claude your control stimulus, treatment stimulus, population description, and outcome variable. Get back a structured design package.
4. **Stage 4 (synthetic test):** pick a total N and batch size (10 is a safe default), run it, get back `subjects_data.csv`.
5. **Stage 5 (analysis):** balance check -> regression -> chart. Read the balance report before trusting the regression.
6. **Stage 7 (writeup):** assemble Method/Balance/Results into a Word doc, optionally benchmarked against a published study.

Worked example at a glance (the workshop's own case study, so you know what a completed run looks like):

Step	What happened
Design	"\$5 a day" (treatment) vs. "\$150 a month" (control) savings framing; primary outcome = sign-up decision
Pilot	N=20, blinded; 80% vs. 40% sign-up, directionally large but only marginal ($p \sim .08$)
Scale-up	N=400 after unblinding; 64.5% vs. 45.5% sign-up, $B=0.82$, $p < .001$
Writeup	Compared against a real published study: savings-decision effect replicated, affordability effect replicated at about half the size, understandability effect did not replicate

That last row is the point worth remembering: a synthetic run can look convincing and still miss part of the real psychology. Treat it as a fast way to iterate on a design, not a replacement for real data.

1. Prerequisites & Setup

This section gets you from a bare machine to a working environment: VS Code, Claude Code, Python, and git/GitHub.

1.1 Install VS Code and Claude Code

1. Install **VS Code** from its official source.
2. Install the **Claude Code** extension (or CLI) and sign in.

3. Open a folder in VS Code -- this becomes your project's working directory.

1.2 Install Python

Check whether Python is already installed:

```
python --version
```

If it's missing (Windows sometimes shows a Microsoft Store stub instead of a real interpreter), install it:

```
winget install Python.Python.3.12
```

Gotcha: after installing anything with winget, PATH changes do not appear in your *current* terminal session automatically. Refresh it manually, or open a new terminal window:

```
$machinePath =  
[System.Environment]::GetEnvironmentVariable("Path","Machine")  
$userPath = [System.Environment]::GetEnvironmentVariable("Path","User")  
$env:Path = $machinePath + ";" + $userPath
```

1.3 Create a project-local virtual environment

```
python -m venv .venv  
.\.venv\Scripts\python.exe -m pip install --upgrade pip  
.\.venv\Scripts\python.exe -m pip install -r scripts/requirements.txt
```

A minimal requirements.txt for this kind of workflow:

```
pandas  
numpy  
scipy  
statsmodels  
matplotlib  
openpyxl  
python-dotenv  
tabulate
```

(Add anthropic only if you plan to call the Claude API directly from a script, rather than using Claude Code subagents.)

1.4 Install git and set up GitHub

```
winget install --id Git.Git  
winget install --id GitHub.cli --source winget --scope user
```

Gotcha: some winget installers try to trigger an admin-elevation (UAC) prompt, which silently fails in a non-interactive terminal. Adding `--scope user` installs to your user profile instead and avoids the prompt entirely.

Authenticate GitHub (this step is interactive -- run it yourself in a real terminal, not through an automation tool):

```
gh auth login
```

Then create and push your repo:

```
git init
git add .
git commit -m "Initial commit"
gh repo create <repo-name> --private --source=. --remote=origin --push
```

1.5 Install pandoc (for generating docx/pptx/pdf output)

```
winget install --id JohnMacFarlane.Pandoc --scope user
```

PDF conversion additionally needs a rendering engine. A lightweight option:

```
winget install --id Typst.Typst --scope user
```

Then convert with:

```
pandoc mydoc.md -o mydoc.pdf --pdf-engine=typst
```

Gotcha: if your markdown embeds images by relative path, pandoc resolves paths relative to your *current directory*, not the markdown file's location. Fix with `--resource-path=". ;path\to\assets"` or by running pandoc from the file's own directory.

2. Project Scaffolding

A recommended directory layout for this kind of workflow:

```
my-research-env/
  .claude/
  skills/
    behavioral-design/SKILL.md
    synthetic-ab-test/SKILL.md
    data-analysis/SKILL.md
    research-writeup/SKILL.md
  scripts/
    requirements.txt
    synthetic_sample/
```

```
sample_population.py
persona_prompt.py
build_batches.py
merge_responses.py
analysis/
  balance_check.py
  regression.py
  charts.py
projects/
  <study-slug>/
    03_design/
      research_design.md
      variables_spec.md
      population_spec.md
      stimuli/
        control.md
        treatment.md
    04_synthetic_test/
      population_config.json
      test_config.json
      subjects_frame.csv
      subjects_data.csv
    05_analysis/
      balance_report.md
      regression_results_*.md
      charts/*.png
    07_writeup/
      writeup.md
      writeup.docx
.gitignore
```

A .gitignore that keeps machine-specific and secret-bearing files out of version control:

```
.venv/
__pycache__/
*.pyc
.env
```

3. Stage-by-Stage Guide

Each stage below is a Claude Code **Skill** -- a markdown instruction file the AI agent follows. You invoke a stage by describing what you want in plain language; Claude Code figures out which skill applies (or you can name it directly, e.g. "run behavioral-design").

3.1 Stage 3 -- Behavioral Design (intake)

What it does: takes your control stimulus, treatment stimulus, population description, and outcome variable(s), and turns them into a structured design package Stage 4 can consume.

Copy-paste prompt template:

I want to set up a behavioral test.

Control condition stimulus: [PASTE CONTROL STIMULUS TEXT]

Treatment condition stimulus: [PASTE TREATMENT STIMULUS TEXT]

Population to sample: [DESCRIBE DEMOGRAPHICS AND INDIVIDUAL DIFFERENCES -- e.g. age mean/SD, gender split, income, education, financial literacy, numeracy -- with means/SDs if you have them, or a rough text description if not]

Outcome variable(s): [NAME AND TYPE -- e.g. "a binary choice between X and Y" or "a 1-7 Likert rating of Z"]

Please set this up as a new project called [PROJECT-SLUG].

What you get back: `research_design.md`, `stimuli/control.md` and `stimuli/treatment.md`, `population_spec.md`, `variables_spec.md` -- review these before moving on, especially `variables_spec.md`'s note on which category should be coded 1 for any binary outcome.

3.2 Stage 4 -- Synthetic A/B Test

What it does: samples a synthetic subject population, randomly assigns them to conditions, and collects each subject's response by simulating them with Claude (subagents, batched) or the direct API.

population_config.json field reference:

Field	Meaning
conditions	list of condition names
n_total	total sample size
allocation	"equal" or a dict of exact per-condition counts
variables	list of {name, kind: continuous\ categorical, role, mean/sd/min/max/round} or {categories, probabilities}
correlations	optional list of {var1, var2, rho} pairs

test_config.json field reference:

Field	Meaning
mode	"subagent" (default, no API key) or "api" (opt-in, needs ANTHROPIC_API_KEY)
conditions	dict of condition name -> stimulus text
outcome_variable	{name, type, options/scale_min/scale_max, description, reasoning_prompt, secondary: [...]}
batch_size	subjects simulated per subagent call (default 10)
agent_type	which subagent type to use (default general-purpose)

Copy-paste prompt template:

Let's run the synthetic A/B test for [PROJECT-SLUG].

Total N: [N]. Allocation: [equal / specify counts per condition].

Use subagent mode with batch size [10 is a good default -- smaller batches are more reliable per-subject, larger batches are more token-efficient].

What you get back: subjects_frame.csv (sampled profiles + condition), subjects_data.csv (profiles + condition + responses) -- spot-check a few rows before moving to analysis.

Concrete example: a filled-in population_config.json

This is an abbreviated version of the config used for the workshop's own pennies-a-day case study (N reduced for space):

```
{
  "conditions": ["condition_1", "condition_2"],
  "n_total": 400,
  "allocation": "equal",
  "variables": [
    {"name": "age", "kind": "continuous", "role": "demographic",
     "mean": 34.29, "sd": 7.24, "min": 18, "max": 85},
    {"name": "gender", "kind": "categorical", "role": "demographic",
     "categories": ["female", "male"], "probabilities": [0.477, 0.523]},
    {"name": "income_bracket", "kind": "continuous", "role":
"demographic",
     "mean": 5.29, "sd": 3.2, "min": 1, "max": 16, "round": true},
    {"name": "financial_literacy", "kind": "continuous", "role":
"individual_difference",
     "mean": 2.36, "sd": 0.91, "min": 0, "max": 3, "round": true}
  ],
  "correlations": []
}
```

And the matching test_config.json:

```
{
  "mode": "subagent",
  "conditions": {
    "condition_1": "...get started with $5 a day today...",
    "condition_2": "...get started with $150 a month today..."
  },
  "outcome_variable": {
    "name": "choice",
    "type": "binary",
    "options": ["Not now", "Yes, start saving"],
    "description": "Whether the subject chooses to start saving.",
    "reasoning_prompt": "List three things you were thinking about, in
your own words."
  },
  "batch_size": 10,
  "agent_type": "general-purpose"
}
```


Notice round: true on income_bracket and financial_literacy -- both are ordinal/count scales in the real instrument (a 16-point income bracket, a 0-3 correct-answer count), so fractional sampled values need rounding to stay meaningful.

3.3 Stage 5 -- Data Analysis

What it does: checks randomization balance, runs regressions on your outcome variable(s), and produces bar charts.

Copy-paste prompt template:

Run the data analysis for [PROJECT-SLUG]:

- Balance check across all sampled demographic/individual-difference variables
- Regression on [OUTCOME VARIABLE NAME], type [binary/continuous], with controls: [none, or list variables]
- If binary, code [WHICH CATEGORY] as 1
- Bar chart of the outcome by condition

What you get back: balance_report.md, regression_results_<outcome>.md, charts/<outcome>_by_condition.png -- read the balance report first; if anything is flagged, consider re-running the regression with that variable as a control.

3.4 Stage 7 -- Research Writeup

What it does: assembles Method, Balance, and Results sections into a Word-ready document, optionally compared against a reference study.

Copy-paste prompt template:

Generate a Stage 7 writeup for [PROJECT-SLUG] (results-and-method scope, no mediation/moderation).

[Optional] Compare against this reference study: [CITATION]. Reference statistics: [OUTCOME NAME: effect size, p-value, with/without controls -- repeat per outcome you have a reference value for].

What you get back: writeup.md and writeup.docx. If you included a reference comparison, read the classification (replicates / partially replicates / does not replicate) critically -- don't let a partial or non-replication get smoothed over in the prose.

4. Critical-Steps Checklist

- **Batch size is a real tradeoff.** Smaller batches (e.g. 2 subjects/call) cost roughly 4x more tokens per subject than larger batches (e.g. 10/call), because fixed prompt overhead (persona instructions, stimuli, schema) gets amortized across more subjects. But very large batches risk subjects' answers drifting toward each other. Start at 10 unless you have a reason not to.
- **Decide your blinding approach before Stage 4.** If you want to avoid analyst bias, ask for neutral condition labels (Condition A/B) and don't reveal which is treatment until after Stage 5 is complete.
- **When balance check flags an imbalance,** don't panic and don't ignore it either -- re-run the regression with that variable as a control and confirm the main result holds. With many variables tested, some false positives are expected by chance (roughly a 1-in-4 chance of at least one $p < .05$ false positive across 6 independent tests at $\alpha = .05$).
- **Binary outcome coding is not automatic-safe.** Don't rely on an alphabetical default for which category is coded 1 -- state it explicitly, since it determines the sign of every downstream effect.
- **Estimate token cost before scaling up.** At batch size 10, expect roughly 2,500-2,600 tokens per synthetic subject in subagent mode. For N subjects, that's very roughly $N \times 2,600$ tokens total, split across $N / \text{batch_size}$ subagent calls.
- **A pilot run (small N) is worth doing before a full-scale run.** It catches configuration and prompt problems cheaply, and gives you a directional read even if it's not statistically definitive on its own.

4.1 Sizing a run: rough token/effort reference

These are rough, empirically-observed figures from subagent mode at batch size 10 -- use them to sanity-check a run's cost/time before launching it, not as exact guarantees.

N (total subjects)	Batches (batch size 10)	Approx. tokens	Notes
20	2	~50,000	good for a quick pilot
100	10	~260,000	moderate live run, still manageable in one session
400	40	~1,040,000	a full-scale run; launch batches in parallel groups (e.g. 6-8 per message) to keep wall-clock time reasonable

For comparison, the same N in **API mode** (opt-in, direct Anthropic API calls, not batched) is roughly 600-800 tokens/subject -- cheaper in raw tokens and billed separately in dollars rather than counting against your Claude Code usage, but requires its own API key and billing setup, and doesn't benefit from batching efficiency.

5. Troubleshooting Appendix

Symptom	Cause	Fix
winget install reports success then the tool still isn't found	PATH updated on disk but not in your current shell process	Refresh <code>\$env:Path</code> from Machine+User scope, or open a new terminal
winget install fails/cancels with an installer exit code around 1602	Installer tried to trigger a UAC elevation prompt that can't be answered non-interactively	Retry with <code>--scope user</code>
pandoc converts markdown but images are missing from the output	pandoc resolves relative image paths from the working directory, not the markdown file's folder	Add <code>resource-path=".;path\to\assets"</code> , or cd into the markdown file's folder first
A regression script errors on a column that "should" be numeric	Newer pandas versions may store text as a <code>str</code> dtype instead of the classic object dtype, breaking <code>dtype == object</code> checks	<code>pd.api.types.is_numeric_dtype(...)</code> instead of comparing dtype directly
A statistical test (e.g. Bartlett's) throws an obscure internal array error	Some scipy versions mishandle integer-dtype input in certain test implementations	Cast the column to float before passing it to the test
A subagent's JSON response has extra text before/after the array	Subagents don't have a forced-output guarantee the way a direct API tool-call does	Instruct explicitly to return <i>only</i> the JSON array with no surrounding text; strip/retry if it still wraps the output in a code fence
git commit succeeds but shows a warning about name/email being auto-detected	git had no configured identity and guessed one from your username/hostname	Set it explicitly: <code>git config --global user.name "..."</code> and <code>git config --global user.email "..."</code> ; amend the commit with <code>--reset-author</code> if needed
gh auth login can't complete inside an automated/non-interactive session. A binary outcome's regression coefficient has the "wrong" sign	The device-code/browser login flow needs a human present. The default coding falls back to alphabetical order when no positive category is specified	Run it yourself in a real terminal window; automation can create/push repos that should count as I explicitly rather than relying on a default stored credentials

6. Glossary & Command Cheat Sheet

Synthetic subject -- an LLM-simulated research participant, given a sampled demographic/individual-difference profile and asked to respond in character.

Batch -- a group of synthetic subjects simulated within a single subagent call, to amortize fixed prompt overhead.

Condition -- one arm of a between-subject experiment (e.g. control vs. treatment).

Subagent -- an independent Claude Code conversation spawned to do a self-contained piece of work (here: simulate one batch of subjects) and return a result.

Blinding -- withholding which condition is "treatment" vs. "control" from the analyst until after data collection, to prevent bias in analysis or interpretation.

Balance check -- a statistical test confirming that random assignment produced similar-looking groups across conditions (age, gender, etc.), which is a precondition for trusting a simple between-condition comparison.

Skill -- a markdown file (SKILL.md) that packages instructions for a recurring task, which Claude Code follows when the task matches.

Allocation -- how a study's total N is split across conditions (e.g. "equal" for an even split, or exact per-condition counts).

Positive-label coding -- the explicit choice of which category of a binary outcome is coded 1 (vs. 0), made explicit rather than left to an alphabetical default so the sign of every effect is interpretable.

Balance imbalance (false positive) -- a statistically significant difference between conditions on a variable that should, in truth, be balanced by randomization; expected to occur by chance some fraction of the time across many tests, so it should prompt a robustness check rather than an assumption of a broken sampler.

Effect size -- the magnitude of a treatment's impact (e.g. a regression coefficient), as distinct from its statistical significance (p-value); a result can be significant with a small effect size, or large but not significant at a small N.

Mediator / moderator (out of scope here) -- a mediator is a variable that explains *why* an effect occurs (e.g. affordability perception explaining a framing effect); a moderator is a variable that changes *how strong* an effect is for different subgroups (e.g. financial

literacy). Both require dedicated analyses (Query Theory / GSEM mediation, floodlight moderation) not included in this workflow's v1 scope.

Replication assessment -- classifying a result against a reference study as *replicates* (same sign, both significant), *partially replicates* (same sign, only one significant), or *does not replicate* (opposite sign, or a near-zero estimate where the reference was significant).

Command cheat sheet

```
# Refresh PATH in a new PowerShell session
$machinePath =
[System.Environment]::GetEnvironmentVariable("Path","Machine")
$userPath = [System.Environment]::GetEnvironmentVariable("Path","User")
$env:Path = $machinePath + ";" + $userPath

# Create and populate a venv
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r scripts\requirements.txt

# Git + GitHub
git init
git add .
git commit -m "message"
gh repo create <name> --private --source=. --remote=origin --push
git push

# pandoc conversions
pandoc doc.md -o doc.docx
pandoc doc.md -o doc.pptx
pandoc doc.md -o doc.pdf --pdf-engine=typst --resource-path=".;assets"
```